

About the Prediction Methodology Standard

What Is the Prediction Methodology Standard?

The Prediction Methodology Standard (PMES) is the Parent Standard within PGA™ — Prediction Governance Architecture that governs the methodology by which a prediction is constructed.

It establishes whether the predictive method is identifiable, structured, documented, applicable to the prediction being made, and sufficiently traceable across methodological changes and versions.

PMES does not decide what the prediction should be.

It governs whether there is a sufficiently defined methodology behind it.

A prediction without a documented methodology is an opinion, not a governed prediction.

Why the Standard Exists

Predictions increasingly influence investment, infrastructure, markets, public policy, organizational planning, risk management, AI systems, autonomous systems, and individual decisions.

Yet a prediction may appear precise without revealing how it was produced.

A percentage, forecast, model output, expert statement, market-implied probability, or AI-generated prediction can create an appearance of methodological authority even when the underlying method is unclear.

PMES addresses this governance problem.

It requires the methodology behind a governed prediction to be sufficiently identifiable and documented so that the prediction can be understood in relation to the method that produced it.

Precision of output does not establish quality of methodology.

What the Standard Governs

PMES governs the Prediction Methodology used to transform a governed predictive frame into a prediction.

At the Parent Standard level, this includes governance of:

predictive method selection; methodological structure; predictive logic; methodological documentation; method applicability; methodological dependencies and limitations; methodology traceability; methodology change; methodology version governance.

The standard is technology-neutral.

The methodology may involve statistical analysis, mathematical models, AI systems, machine learning, human judgment, expert forecasting, prediction markets, ensembles, simulation-supported processes, or hybrid approaches.

What matters is not which method is used.

What matters is whether the method behind the prediction can be methodologically governed.

The Method Must Be Distinguishable from the Prediction



A prediction is an output about an unresolved or future condition.

A Prediction Methodology is the structured mechanism through which that output is produced.

These must remain separate.

Prediction Methodology $\neq$ Prediction

Prediction Model $\neq$ Prediction Methodology

Model Output $\neq$ Governed Prediction

A sophisticated model may form only one component of a larger predictive methodology.

Likewise, a simple methodology may be legitimate if it is appropriate, sufficiently defined, and correctly bounded.

Methodology Must Fit the Prediction

A method cannot be considered appropriate merely because it is powerful, popular, historically successful, institutionally accepted, or technically capable of generating an output.

Its applicability must be considered in relation to the Prediction Object, Predictive Purpose, Prediction Target, Prediction Horizon, and Predictive Boundary established earlier in PGA™.

Method Availability $\neq$ Method Applicability

Predictive Capability $\neq$ Methodological Suitability

A methodology capable of predicting one type of outcome may be inappropriate for another.

Methodological Documentation

PMES requires sufficient documentation to establish what methodology governed the prediction.

The requirement is not documentation for its own sake.

Documentation must preserve enough methodological information to understand the relevant structure, logic, applicability, limitations, dependencies, and version of the method used.

This is particularly important when predictions are later communicated, revised, compared with outcomes, or evaluated for performance.

Documentation Volume $\neq$ Methodological Governance Quality

Human, AI, and Hybrid Prediction

PGA™ does not assume that human predictions are inherently less governable than machine predictions, or that machine-generated predictions are inherently more rigorous.

A prediction may originate from:

Human Forecaster

Expert Group

Statistical Model

AI System

Prediction Market

Model Ensemble

Human–AI System

Hybrid Predictive Methodology

The governance question remains the same:

By what documented and reproducible methodology is the prediction legitimately constructed?

Probability Must Have Methodological Provenance

Two predictions may both state a probability of 70% while deriving that number through completely different mechanisms.

One may originate from a statistical model.

Another may represent expert judgment.

Another may be generated by an AI system.

Another may represent a prediction-market price.

Another may result from an ensemble or hybrid methodology.

The numerical value alone does not establish methodological equivalence.

Same Probability Value $\neq$ Same Predictive Meaning

PMES preserves the methodological origin of the probability-like output. Later PGA™ governance determines how probability and uncertainty may legitimately be represented.

Methodology Must Remain Historically Identifiable

Prediction methodologies evolve.

Models are retrained. Algorithms change. Expert procedures are revised. Data-processing methods change. Ensemble composition changes. AI systems are replaced.

A later methodological improvement must not silently become the methodology attributed to an earlier prediction.

Current Methodology $\neq$ Historical Methodology

Methodology Update $\neq$ Historical Replacement

PMES therefore establishes methodology version governance as part of Prediction Methodology governance.

Position Within PGA™

PMES follows establishment of the governed predictive frame under POPS — Prediction Object & Purpose Standard .

The complete PGA™ Governance Flow is:

Prediction Object & Purpose → → Prediction Methodology → Prediction Basis → Prediction Assumptions → Prediction Formation → Probability & Uncertainty → Prediction Feasibility → Collective Prediction where applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance → Prediction Record & Continuity

Current Position: Prediction Methodology

PMES establishes how the prediction is intended to be methodologically constructed before PGA™ governs the information and signals forming its Prediction Basis.

How PMES Is Structured

PMES is operationalized through five Core Modules:

PMSM — Predictive Method Selection Module Governs selection of the predictive method appropriate to the governed predictive frame.

MSDM — Methodological Structure Definition Module Governs the structure and relationships through which the selected methodology operates.

PLDM — Predictive Logic Documentation Module Governs documentation of the material logic connecting predictive conditions to predictive output.

MAPM — Method Applicability Module Governs whether the methodology is applicable within the predictive context in which it is intended to operate.

MVGM — Methodology Version Governance Module Governs methodology versions, material methodological changes, and historical methodological continuity.

Together they establish the governed Prediction Methodology.

What PMES Does Not Govern

PMES does not determine:

which information constitutes sufficient Prediction Basis; whether evidence itself is valid; what predictive assumptions should hold; the specific prediction produced; probability or uncertainty representation; prediction feasibility; collective prediction legitimacy; permitted reliance; Operational Reality; whether the prediction ultimately proves correct.

Those questions belong to subsequent PGA™ Parent Standards or other applicable OOF® architectures.

PMES governs the methodology behind the prediction — not the future the prediction describes.

Relationship to Other OOF® Architectures

Where simulation contributes to prediction, SIMULOS® continues to govern simulation.

Where Operational Reality is later established, ORA™ retains governance ownership of Operational Reality.

Where evidence requires governance, ADIT® retains evidence-governance ownership.

Where validation is required, VALIDOS™ governs validation.

Where AI, cognition, autonomous operation, accountability, or integrity are involved, their respective OOF® architectures retain their governed domains.

PGA™ governs how relevant outputs participate in the Prediction Lifecycle without absorbing the governance spaces that produced them.

Why the Standard Matters

Modern systems can generate enormous numbers of predictions.

Generating an output is becoming easier.

Establishing why that output should be treated as a methodologically governed prediction remains a separate problem.

Without methodology governance, organizations may compare predictions that were produced under fundamentally different methods, rely upon probabilities whose methodological origin is unclear, evaluate historical performance without knowing which methodology version produced the result, or treat model output as though the existence of the model itself established predictive legitimacy.

PMES creates the methodological reference layer required to prevent these failures.

Use Case 1 — High-Stakes Prediction Without an Explainable Methodological Basis

An organization uses predictions to support consequential decisions involving capital allocation, infrastructure, operational capacity, safety, public resources, or strategic planning.

The predictive system repeatedly produces precise numerical forecasts and probability estimates.

The organization can see the outputs, but cannot establish clearly which predictive method produced them, why that method was selected, how the relevant predictive logic operates, whether the method is appropriate for the Prediction Object and Horizon, or which methodology version produced a historical prediction.

The central governance problem is therefore not initially whether the prediction is correct.

It is:

Can an organization legitimately rely upon a prediction when it cannot establish the methodology under which that prediction was produced?

PMES establishes the methodological governance necessary to answer that question before predictive accuracy alone is allowed to substitute for methodological legitimacy.

Use Case 2 — Two Predictions, Same Probability, Different Methodological Meaning

Two predictive systems independently state:

70% probability that Outcome X will occur.

The numerical outputs appear identical.

The first prediction is produced through a documented statistical methodology using defined historical inputs.

The second is derived from a prediction market whose price reflects participants with different information access, incentives, capital influence, and predictive relationships to the outcome.

A third system could potentially generate the same 70% through AI inference or expert judgment.

The governance problem is:

Can identical probability values be treated as equivalent predictions when the methodologies that produced them are materially different?

PMES establishes that the methodology and methodological provenance behind each prediction must remain identifiable.

The probability value alone cannot establish methodological equivalence.

Same Probability Value $\neq$ Same Predictive Meaning

Canonical Boundary

PMES governs the selection, identity, structure, documentation, applicability, methodological provenance, limitations, change, versioning, and continuity of the methodology used to construct a prediction. It does not govern Prediction Basis sufficiency, assumptions, prediction formation, probability and uncertainty representation, feasibility, collective prediction legitimacy, reliance, Operational Reality determination, predictive performance, or Prediction Record continuity.

Canonical Closing Statement

The ability to generate a prediction does not establish the methodology necessary to govern it.

A prediction without a documented methodology is an opinion, not a governed prediction.

Governed Links

[About Prediction Methodology Standard \(PMES\)](#)

[Prediction Methodology Specification Module \(PMSM\)](#)

[Methodological Structure Definition Module \(MSDM\)](#),
[Prediction Logic Documentation Module \(PLDM\)](#),
[Methodology Application Module \(MAPM\)](#),
[Methodology Version Governance Module \(MVGM\)](#).

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